

MARVIN WONG

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EDUCATION

University of California, Los Angeles — M.Eng., Data Science GPA: 4.0 Neural Networks & Deep Learning, NLP, Large-Scale Machine Learning	Sep 2025 – Dec 2026 Los Angeles, CA
University of California, Los Angeles — B.S., Data Theory & Atmospheric/Oceanic Sciences GPA: 3.83 Statistical Modeling, Machine Learning, Data Structures & Algorithms, Probability	Sep 2021 – Jun 2025 Los Angeles, CA

TECHNICAL SKILLS

- **Languages & Platforms:** Python, SQL, R, Git/GitHub, Docker, AWS, Spark, Databricks
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, XGBoost, FAISS, Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, MCP
- **Data & Visualization:** Pandas, NumPy, Tableau, Matplotlib, Excel, ETL

RELEVANT EXPERIENCE

Graduate Research Assistant UCLA Computer Science Department	Mar 2026 – Present Los Angeles, CA
• Improved robotics model task success rate by 20 percentage points (70% → 90%) by training and benchmarking two Behavior Cloning Transformer models in PyTorch across the robomimic and LIBERO benchmarking frameworks • Replaced a time-based approach with a spatial pattern-matching Python retrieval method, producing more consistent, higher-quality training data • Cut a key error metric by over 90% by designing robustness experiments (controlled noise, environment perturbations) and analyzing HDFS results with Python/Matplotlib	
Research Assistant — Atmospheric Data Scientist Hong Kong University of Science and Technology	Jun 2024 – Sep 2024 Hong Kong
• Improved climate forecasting model accuracy by 50% over baseline by developing a time-series (TDA) forecasting model for climate signals • Built an automated Python ingestion pipeline (Pandas, Scikit-learn) that scraped and consolidated 3+ GB of NOAA climate data, streamlining analysis for the research team	
Data Scientist Intern PricewaterhouseCoopers	Jul 2023 – Sep 2023 Hong Kong
• Reduced order-to-submission time by 15% by designing and executing A/B tests (t-tests) on 3 UI changes across ~1,000-user test groups • Prevented critical defects from reaching production by executing UAT across 20 test cases (5–6 defects found); built executive Tableau dashboards presented to cross-functional stakeholders • Queried procurement system data using SQL from a SAS-hosted platform to support experiment design and downstream statistical analysis • Conducted 4 company-wide training sessions on the redesigned procurement system UI, achieving an 80% switchover rate	
Data Analyst Intern Hysan Development Company	Jun 2022 – Sep 2022 Hong Kong
• Cut reconciliation time by 2 hours/week by engineering a stock ledger system using VLOOKUP for automated data validation and Pivot Tables for dynamic reporting • Modeled foot traffic and revenue scenarios in Excel for a proposed retail development, translating findings into a data-storytelling presentation for non-technical stakeholders	

PROJECTS

Securiport Visual Intelligence Capstone	Jun 2026 – Present
• Built an end-to-end computer vision pipeline in Python (YOLOv8 detection + multi-object tracking) to detect and follow a target individual across video footage from a single reference image • Achieved 98.4% top-1 identification accuracy by engineering an ArcFace-based identity verification system (ONNX Runtime, GPU-accelerated) aggregating cross-frame cosine-similarity scores; layered a Qwen3-VL vision-language model with custom prompt engineering to classify behavior at 96.4% accuracy	
Sci-Fi/Fantasy Book Recommendation Engine	
• Built a two-stage recommendation pipeline (PyTorch, XGBoost) combining behavioral, content-based, and collaborative signals, achieving 0.291 HR@10 / 0.1244 NDCG@10; deployed live on Google Cloud • Drove a 3x improvement in recommendation quality by diagnosing data leakage and pipeline bugs; processed tens of millions of records with PySpark and added CI/CD (GitHub Actions) test coverage	
AI Football Scouting Platform	
• Developed a RAG-based scouting assistant (LangChain, Google Gemini) over a vector database to search narrative scouting reports, backed by a SQL-consolidated player-metrics pipeline enabling instant multi-condition filtering in an interactive Streamlit dashboard	

ACCOMPLISHMENTS

- **2023 American Statistical Association DataFest** — Top 12 Finalist for NLP sentiment analysis and statistical regression on American Bar Association appointment transcripts (Python)